

Automated Detection of Malaria-Infected Cells Using Convolutional Neural Networks

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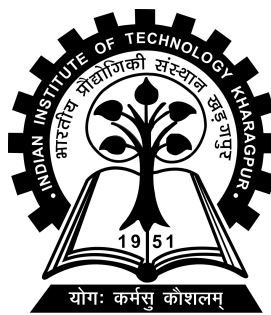
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Autumn Semester, 2024-25

November 2024

DECLARATION

I certify that

- (a) The work contained in this report has been done by me under the guidance of my supervisor.
- (b) The work has not been submitted to any other Institute for any degree or diploma.
- (c) I have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.
- (d) Whenever I have used materials (data, theoretical analysis, figures, and text) from other sources, I have given due credit to them by citing them in the text of the thesis and giving their details in the references. Further, I have taken permission from the copyright owners of the sources, whenever necessary.

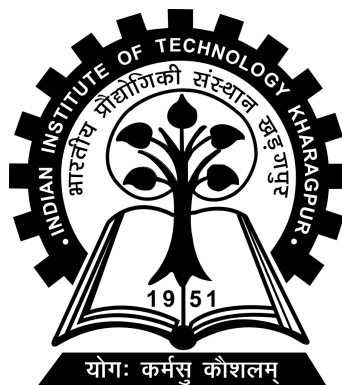
Date: November 2024

Place: Kharagpur

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CERTIFICATE

This is to certify that the project report entitled “Automated Detection of Malaria-Infected Cells Using Convolutional Neural Networks” submitted by Raghav Agarwal (Roll No. 21BT3EP20) to Indian Institute of Technology Kharagpur towards partial fulfilment of requirements for the award of degree of Dual Degree (Btech+Mtech) in Biotechnology & Biochemical Engineering/Engineering Entrepreneurship is a record of bona fide work carried out by him under my supervision and guidance during Autumn Semester, 2024-25.

Date: November 2024

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Abstract

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Department: **Department of Bioscience and Biotechnology**

Thesis title: **Automated Detection of Malaria-Infected Cells Using Convolutional Neural Networks**

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Malaria, a life-threatening disease affects over 200 million people globally each year, with the majority of cases concentrated in low-resource settings. Rapid and accurate diagnosis is crucial for effective treatment and control, but conventional diagnostic methods face limitations in accessibility, cost, and consistency. This thesis explores an automated, machine learning-based approach to malaria identification that combines image segmentation and deep learning classification. Blood sample images are captured under a microscope, and Red Blood Cells (RBCs) are segmented, isolating them for individual analysis. A Convolutional Neural Network (CNN) model, then classifies each RBC as infected or uninfected. This CNN-based method demonstrates the potential of deep learning in the field of digital pathology, opening pathways for future advancements in automated disease diagnosis and expanding the applicability of such networks across various pathological applications.

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Chapter 1

Introduction

Malaria is a life-threatening disease caused by protozoan parasites of the *Plasmodium* genus, which are transmitted through the bites of infected female *Anopheles* mosquitoes. These parasites primarily target red blood cells, leading to a range of symptoms, including fever, fatigue, headaches, and in severe cases, seizures and coma. Malaria is particularly deadly among children in sub-Saharan Africa, where nearly one child dies from the disease every minute, making it one of the leading causes of childhood neuro-disability.

Malaria remains a critical public health issue, particularly in regions where access to effective healthcare and diagnostic tools is limited. The disease's symptoms can often be confused with those of other illnesses, making early detection challenging. The global burden of malaria is compounded by inadequate diagnostic infrastructure, especially in remote and underserved regions, where traditional diagnostic methods are either unavailable or too costly. Therefore, innovative solutions to enhance the speed and accuracy of malaria detection are crucial to curbing the disease's impact.

Traditional diagnostic methods, including microscopic examination of stained blood smears, are widely used to identify *Plasmodium*-infected RBCs. However, these

methods are labor-intensive and require skilled personnel, posing challenges in resource-limited settings. Advances in technology, particularly in machine learning and image processing, offer innovative solutions to address these limitations. Automation in malaria diagnosis promises not only to enhance the efficiency of detection but also to provide consistent and reliable results, minimizing the risks associated with human error.

Automated detection methods using Convolutional Neural Networks (CNNs) have shown significant potential in improving malaria diagnostics. By analyzing blood smear images, CNNs can efficiently differentiate between infected and uninfected RBCs, enabling rapid and accurate identification. This approach not only reduces the dependency on expert microscopists but also minimizes human error, making it particularly beneficial in regions with high disease prevalence. Moreover, CNNs can be trained to recognize subtle patterns in blood smears that may be overlooked by human examiners, thus improving detection sensitivity and specificity.

In addition to CNN-based techniques, the integration of image segmentation algorithms plays a key role in isolating individual cells for classification. Segmentation allows for more precise identification of infected cells by breaking down complex images into smaller, manageable units. This improves the accuracy of the subsequent classification step and allows for better handling of images with varying quality and resolution, which are common in field conditions. By segmenting the microscope images into individual RBCs, this project aims to facilitate the development of a fully automated system that can classify each cell as either infected or uninfected.

The present work, explores the application of machine learning techniques, particularly CNNs, for the detection and classification of malaria-infected red blood cells. The project focuses on enhancing the accuracy of segmentation algorithms and improving the overall workflow from image acquisition to final diagnosis. By leveraging existing image datasets and custom-built models, the research aims to develop an efficient and scalable solution to malaria detection.

Given the global burden of malaria and the need for accessible and efficient diagnostic tools, integrating CNN-based techniques into diagnostic workflows can revolutionize how malaria is detected and managed. These advancements have the potential to save countless lives, particularly in high-risk populations and underserved regions. As part of this project, the focus on improving the accuracy, speed, and accessibility of malaria diagnostics serves as a step toward addressing this pressing global health challenge.

Chapter 2

RELATED WORK

In recent years, significant progress has been made in the field of malaria detection and red blood cell (RBC) segmentation, driven by advancements in image processing and machine learning techniques. Researchers have explored various methods, including traditional image processing, classical machine learning algorithms, and deep learning models, to segment RBCs and detect malaria-infected cells.

2.1 Classical Machine Learning Methods

Earlier approaches to malaria detection relied heavily on traditional image processing techniques, such as thresholding, edge detection, and morphological operations, to segment RBCs and identify malaria-infected cells. These methods, while effective for certain image conditions, struggle to generalize across diverse datasets, particularly when the images exhibit variations in cell shapes, sizes, and lighting conditions. Machine learning models, such as support vector machines (SVMs), k-nearest neighbors (KNN), and decision trees, were then applied to classify cells based on extracted

features. However, these models often require manual feature extraction and struggle with high-dimensional data, leading to the adoption of more sophisticated deep learning methods.

2.2 Deep Learning Approaches

Deep learning techniques, particularly convolutional neural networks (CNNs), have emerged as the state-of-the-art solution for malaria detection and RBC segmentation. CNNs excel at automatically learning hierarchical features from raw image data, eliminating the need for manual feature engineering and improving accuracy. Several studies have demonstrated the power of CNNs for RBC segmentation and malaria detection. For instance, a study by MDPI (2020) used a lightweight CNN model to achieve 97% per-pixel segmentation accuracy for RBCs. The model was designed for real-time analysis, making it suitable for deployment on less powerful devices, which is a crucial consideration for mobile or field-based diagnostics.

Pre-trained architectures such as VGG16, ResNet50, and InceptionV3 have been employed for transfer learning, leveraging large-scale datasets to fine-tune the models for specific malaria detection tasks. These architectures have shown high performance, especially when trained on high-resolution datasets. However, studies have also demonstrated that simpler CNN models with fewer layers, such as shallow networks with 2-3 convolutional layers, can achieve comparable or even superior results (95-99% accuracy) with data augmentation techniques. These lighter models are computationally less expensive, making them ideal for environments where hardware resources are limited.

2.3 Hybrid and Advanced Models

Beyond standard CNNs, more sophisticated architectures have been explored to improve the accuracy of malaria detection and RBC segmentation. For example, Faster R-CNN, which combines region proposal networks (RPN) with object detection capabilities, has been applied to identify and classify RBCs in malaria images. This model can efficiently segment both infected and non-infected cells by first proposing regions of interest (ROIs) and then classifying them. However, Faster R-CNN models can face difficulties with the accurate classification of different stages of malaria infection, especially when dealing with imbalanced datasets or subtle morphological changes in infected RBCs. These challenges highlight the need for fine-tuning, advanced training strategies, and more comprehensive labeled datasets to enhance model precision.

Researchers have also explored combining CNNs with other techniques such as U-Net for segmentation tasks. U-Net, originally designed for biomedical image segmentation, is particularly effective in preserving spatial information, making it a powerful tool for RBC segmentation in blood smear images. The encoder-decoder architecture of U-Net allows it to capture both low-level and high-level features, which is crucial for accurate segmentation of cells in highly variable images.

Chapter 3

Dataset and Experimental Setup

3.1 Dataset Overview

The study utilized two datasets tailored for distinct yet interconnected tasks of red blood cell (RBC) segmentation and classification. For segmentation, a custom dataset comprising 706 high-resolution blood smear images containing over 20,000 labeled RBCs across 12 classes was employed. The images, captured under a high-magnification microscope, provided the fine-grained details essential for precise segmentation. Each RBC was annotated with x, y coordinates and labeled with its type, offering a comprehensive ground truth for model training. This dataset was split into 80% training and 20% validation subsets, with data augmentation techniques like random rotations, flips, and intensity normalization applied to improve model generalization. Figure 3.1 shows a sample of the dataset, highlighting the diversity and detail of the annotated RBCs.

For classification, the National Library of Medicine (NLM) malaria dataset was used, containing labeled RBC samples as either malaria-infected or non-infected. This publicly available dataset ensured diversity in cell morphology and staining variations, making it suitable for training the classification model. The data was

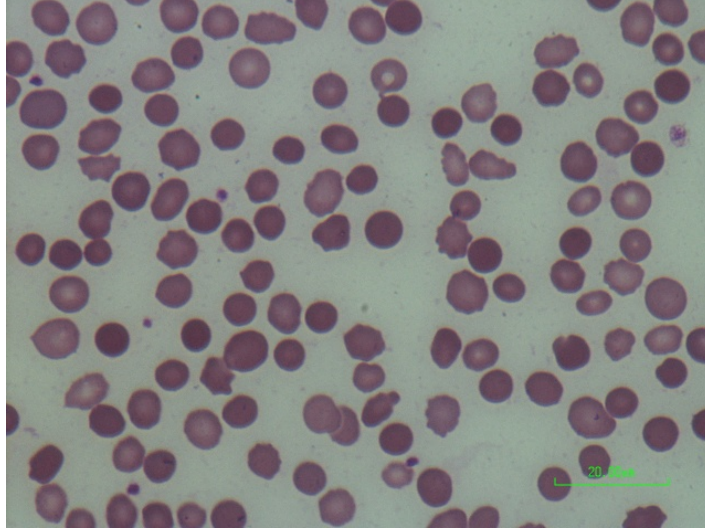


FIGURE 3.1: Sample Annotated RBCs from High-Resolution Blood Smear Images

also divided into 80% training and 20% validation subsets, with preprocessing steps like normalization and augmentations to improve generalization. Figure 3.2 shows a sample of malaria-infected red blood cells, while Figure 3.3 illustrates a sample of non-infected red blood cells, highlighting the visual differences between the two categories.

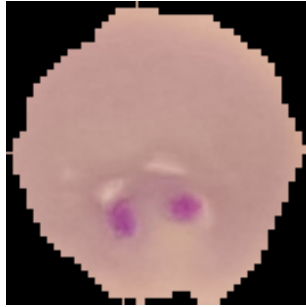


FIGURE 3.2: Malaria-infected RBC

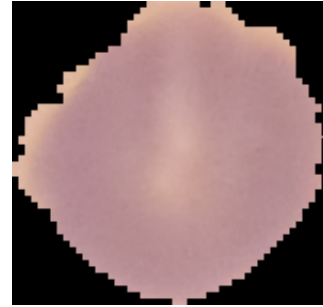


FIGURE 3.3: Non-infected RBC

3.2 Segmentation Model

The segmentation task employed the U-Net architecture, a convolutional neural network (CNN) specifically designed for biomedical image segmentation. U-Net's

encoder-decoder structure allowed for the efficient capture of contextual information while preserving spatial resolution, which was crucial for accurately segmenting overlapping or clustered RBCs. Skip connections between corresponding encoder and decoder layers helped retain fine details.

3.2.1 Key configurations

- **Input resolution:** Images were resized to 512x512 pixels to capture finer details for improved segmentation accuracy.
- **Loss function:** A Dice coefficient loss was utilized to optimize segmentation performance by balancing false positives and false negatives.
- **Optimizer:** The Adam optimizer with a learning rate of 0.0005 was employed, offering faster convergence and adaptive learning rate adjustments.
- **Batch size:** 32 images per batch were processed, optimizing memory usage for higher resolution inputs.
- **Training setup:** The model was trained for 100 epochs, and validation performance was monitored every 20 iterations to ensure early identification of overfitting.

The segmentation model exhibited high accuracy, effectively separating RBCs from complex backgrounds. U-Net's skip connections and tailored preprocessing techniques contributed significantly to its robust performance.

3.3 Classification Model

The classification task used a custom CNN architecture, designed to distinguish between malaria-infected and non-infected RBCs. The model consisted of multiple

convolutional layers for feature extraction, pooling layers for dimensionality reduction, and fully connected layers for classification. A softmax activation function in the output layer enabled binary classification.

3.3.1 Key configurations

- **Input resolution:** Each cell image was resized to 128x128 pixels to reduce computation while preserving essential features.
- **Loss function:** Binary cross-entropy was employed to compute the error for classifying malaria-infected and non-infected cells.
- **Optimizer:** RMSprop was used with a learning rate of 0.001, providing effective handling of vanishing gradients and better performance for small datasets.
- **Batch size:** 64 images per batch were used to balance memory efficiency and model convergence speed.
- **Training setup:** The model was trained for 75 epochs, with validation performance tracked every epoch for iterative fine-tuning.

The classification model demonstrated strong generalization capabilities, effectively identifying malaria-infected cells, with insights from the training process guiding hyperparameter optimization.

Chapter 4

Results

4.1 Segmentation Performance

The segmentation model demonstrated strong performance, accurately isolating red blood cells (RBCs) from the surrounding background. The overall segmentation accuracy was 93.72%, indicating that the model is proficient in distinguishing RBCs in most cases.

The model's ability to match segmented regions with ground truth labels was reflected in the high Jaccard similarity coefficient values. Most segmentations exceeded a coefficient of 0.85, indicating excellent agreement between the predicted and ground truth segments. The coefficients ranged from 0.80571 to 0.94751, showcasing consistent performance across the test set.

Despite these positive results, the model occasionally faced challenges. The central regions of RBCs were sometimes misclassified as background, leading to false negatives. Additionally, platelets and white blood cells, which visually resemble RBCs, were sometimes incorrectly identified as RBCs, leading to false positives.

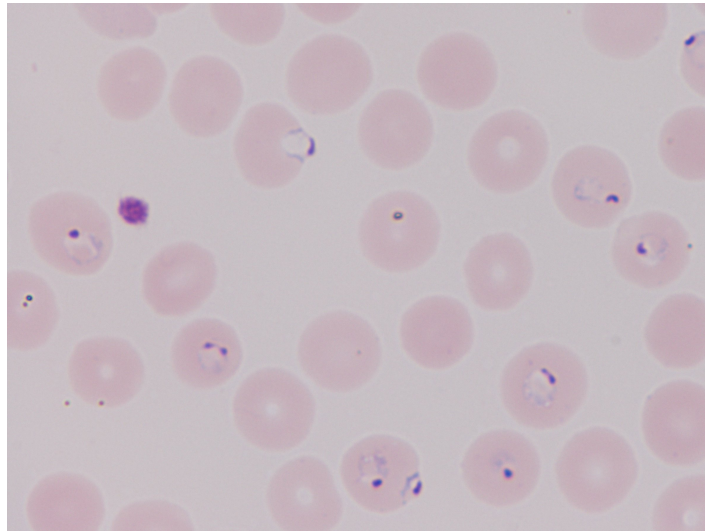


FIGURE 4.1: Original Blood Smear Image

A sample image was processed, and the model successfully segmented 14 individual RBCs, which are displayed in a grid (4.2). These segmentations highlight the model's capacity to isolate RBCs from complex blood smear images, though some areas still require improvement.

4.2 Classification Performance

The classification model performed exceptionally well on the validation set, achieving 95% accuracy, which suggests that the model is capable of distinguishing between infected and non-infected RBCs under balanced conditions. However, performance on the test set was lower, with an accuracy of 75.39%. This drop in accuracy can primarily be attributed to the imbalance in the test dataset, which contained a significantly higher proportion of non-infected RBCs (approximately 4 non-infected to 1 infected RBC).

Key performance metrics for the test set include:

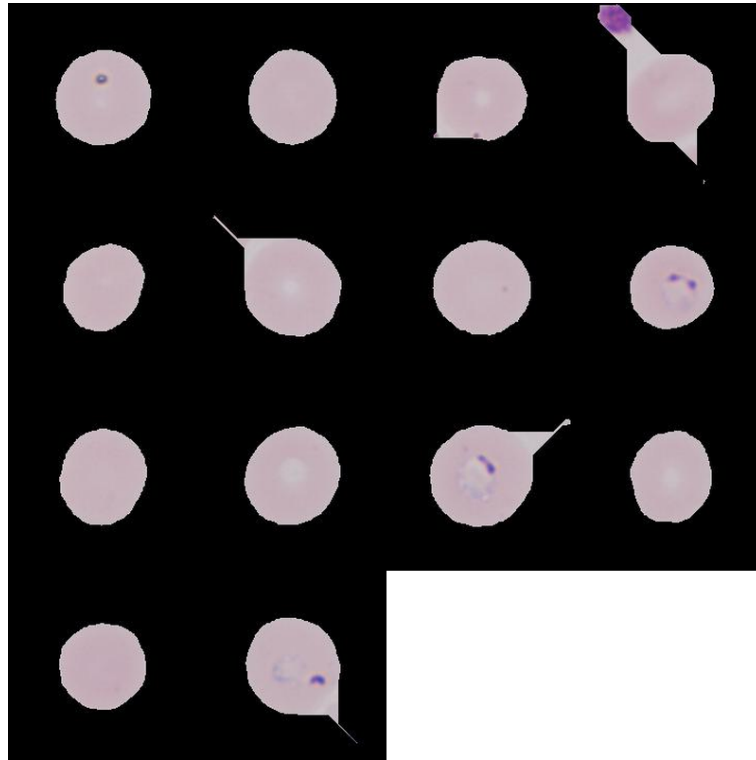


FIGURE 4.2: Segmented RBCs

- **Sensitivity (Recall):** 17.90%, which indicates the model's difficulty in correctly identifying infected RBCs. The low sensitivity suggests that many infected RBCs were missed during classification.
- **Specificity:** 87.04%, reflecting the model's ability to accurately identify non-infected RBCs. This shows that the model is more reliable in identifying healthy cells compared to infected ones.
- **Positive Predictive Value (PPV):** 22.85%, indicating the proportion of predicted infected cells that were actually infected. While the model does identify infected RBCs, many predicted infected cells were false positives.
- **Batch size:** 83.96%, indicating a high rate of correct predictions for non-infected cells.

A major challenge identified during the classification process was the misclassification of platelets as infected RBCs, leading to increased false positives. This issue contributed to the low PPV and sensitivity scores.

A grid (4.3) displays the classification results for the 14 segmented RBCs shown in 4.2, with each RBC labeled as either infected or non-infected. The classification outcomes illustrate the model's current limitations, particularly with regard to distinguishing infected RBCs from non-erythrocytes.

4.3 Application and Visual Analysis

The model was tested on a dataset of 52 high-resolution blood smear images, each measuring 2400x1800 pixels. This dataset, available at Mendeley Data, provided a diverse set of images for the model to process. The RBCs were segmented into individual images of 181x181 pixels, resulting in a total of 551 individual RBCs, with an average of 11 RBCs segmented per image.

For a selected image from the dataset:

- **4.1:** shows the original blood smear image, illustrating the complexity of the sample and the range of RBCs within it.
- **4.2:** presents a grid of the 14 segmented RBCs from the original image, demonstrating the model's ability to isolate individual cells.
- **4.3:** shows the classification results for each segmented RBC, where the cells are labeled as either infected or non-infected. This visual analysis underscores the model's challenges with classifying infected RBCs, particularly in the presence of visual similarities with non-erythrocytes.

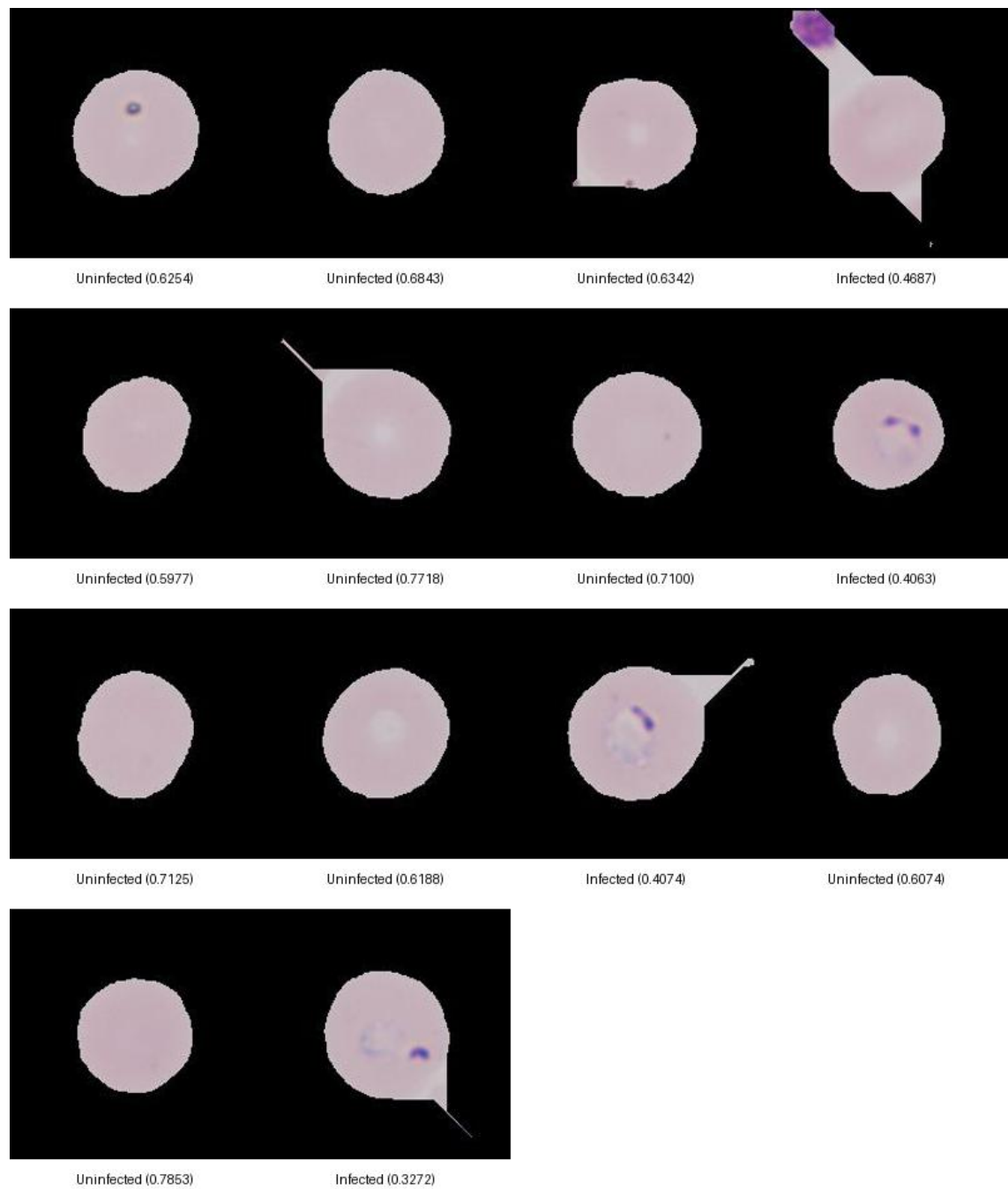


FIGURE 4.3: Classification Results

This section highlights both the strengths and limitations of the model in real-world applications, as well as its potential for future improvement in classifying and segmenting RBCs.

4.4 Malaria Diagnosis Thresholds

Malaria diagnosis involves microscopic analysis of blood smears to determine the proportion of infected RBCs. The World Health Organization (WHO) provides guidelines for malaria diagnosis based on parasitemia levels:

- **Asymptomatic Carriers:** Parasitemia below **5%** without symptoms.
- **Mild Malaria:** Parasitemia between **5-50%**, often with fever and other mild symptoms.
- **Severe Malaria:** Parasitemia exceeding **50%**, associated with symptoms like anemia, organ failure, or cerebral complications.

Stages of Malaria

- **Trophozoite Stage:** Early detection phase; rings in RBCs are visible.
- **Schizont Stage:** High parasitemia levels; indicative of advanced infection.
- **Gametocyte Stage:** Transmission phase; typically detected in severe cases or chronic carriers.

(Reference: *WHO Malaria Guidelines, 2023*).

4.5 Summary of Findings

The segmentation model successfully separated RBCs with high accuracy, and the classification model achieved strong performance on balanced datasets. However, challenges with test set generalization highlight the need for improved training with diverse, imbalanced datasets. The results demonstrate the potential of the integrated approach for real-time malaria diagnostics, particularly when paired with robust preprocessing and augmentation strategies.

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